

■ FindRoot

`FindRoot[lhs==rhs, {x, x0}]` searches for a numerical solution to the equation $lhs==rhs$, starting with $x=x_0$.

`FindRoot[lhs==rhs, {x, {x0, x1}}]` searches for a solution using x_0 and x_1 as the first two values of x . This form must be used if symbolic derivatives of the equation cannot be found. ■ `FindRoot[lhs==rhs, {x, xstart, xmin, xmax}]` searches for a solution, stopping the search if x ever gets outside the range $xmin$ to $xmax$.

■ `FindRoot[{eqn1, eqn2, ...}, {x, x0}, {y, y0}, ...]` searches for a numerical solution to the simultaneous equations eqn_i . ■ `FindRoot` returns a list of replacements for $x, y, \dots$, in the same form as obtained from `Solve`. ■ If you specify only one starting value of x , `FindRoot` searches for a solution using Newton's method. If you specify two starting values, `FindRoot` uses a variant of the secant method. ■ The following options can be given:

<code>AccuracyGoal</code>	6	the accuracy sought in the value of the function at the minimum
<code>MaxIterations</code>	15	maximum number of iterations used
<code>WorkingPrecision</code>	<code>Precision[1.]</code>	the number of digits used in internal computations

■ If `FindRoot` does not succeed in finding a solution to the accuracy you specify within `MaxIterations` steps, it returns the most recent approximation to a solution that it found. You can then apply `FindRoot` again, with this approximation as a starting point. ■ See page 478. ■ See also: `NRoots`, `Solve`.